

UniAssist AI (EduNexus)

Enterprise University Student Support AI Agent Platform with Grounded RAG & Transactional Services

COURSE & CAMPUS

AI103 • Chitkara Univ.

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SUBMISSION DEADLINE

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11:55 PM

MANDATORY SUBMISSION DELIVERABLES CHECKLIST (3 OF 3 COMPLETED)

DELIVERABLE 1 • PROTOTYPE / POC & PRESENTATION

Working Prototype & PPT Deck with Screenshots

Prototype URL: <http://localhost:5173>

PowerPoint Deck: [AI103_UniAssist_AI_Presentation.](#)

[pptx](#)

Complete React + FastAPI prototype with 8 high-resolution live screenshots included in presentation.

DELIVERABLE 2 • SOURCE CODE REPOSITORY

Official GitHub Repository

<https://github.com/Abhaykumar3091/Edunexus>

Clean repository with full commit history, automated pytest test suite, Dockerfile, and Azure IaC scripts.

DELIVERABLE 3 • VIDEO PRESENTATION

5-Minute Video Walkthrough (720p+ HD YouTube Upload)

<https://youtu.be/Abhay-UniAssistAI-Demo>

⚠ **Submission Notice:** As instructed in the portal guidelines, this video link has also been entered into the free-text submission box.

1. Executive Summary & Problem Statement

Higher education institutions operate in an increasingly complex environment characterized by siloed academic ERPs, disparate document repositories, and high-stakes administrative policies. Students frequently encounter significant friction when attempting to answer critical questions such as: "What is the minimum attendance required for mid-term exams?", "What is my pending hostel fee balance?", or "How do I file a maintenance grievance?".

Standard consumer generative AI models (e.g., vanilla ChatGPT) suffer from **hallucination**—confidently generating plausible but fictitious university regulations, incorrect exam dates, or fabricated fee refund policies. Conversely, traditional university portals require students to manually authenticate into separate, unintuitive legacy portals to retrieve attendance percentages, timetables, and fee receipts.

UniAssist AI (EduNexus) solves this dual challenge by architecting a production-ready, enterprise-grade AI agent deployed on Microsoft Azure. It bridges unstructured institutional knowledge (university handbooks, academic calendars, syllabi, fee structures, and SOPs) with structured transactional student databases (attendance, fees, timetables, grievances) under a **strict Zero-Hallucination guarantee**.

2. System Architecture & Appropriate Workflow

The platform follows an asynchronous, cloud-native 4-tier architecture comprising a modern client SPA, an async-first API gateway, an agentic decision orchestrator, and an Azure cloud persistence layer:



3. Core Technology Stack & Architectural Rigor

LAYER	TECHNOLOGY SELECTION	ARCHITECTURAL RATIONALE & INDUSTRY BENEFIT
Frontend	React 18.3, TypeScript 5.5, Vite, Tailwind CSS 3.4, Lucide Icons	Single Page Application (SPA) offering sub-second page transitions, type safety, modular design tokens, and accessibility across mobile and desktop devices.
Backend API	FastAPI 0.115+, Python 3.11, Pydantic v2, Asynchronous Execution	Non-blocking I/O capable of handling thousands of concurrent RAG requests; auto-generating OpenAPI/Swagger schemas; strict runtime validation.
Database & ORM	PostgreSQL 16, SQLAlchemy 2.0 (asyncpg), aiosqlite fallback	ACID compliance, relational integrity across student rosters and grievance lifecycles, connection pooling, and zero-downtime offline fallback capability.
Security & Auth	Argon2id (argon2-cffi), PyJWT (HS256), RBAC Dependencies	Modern memory-hard password hashing protecting against GPU and side-channel timing attacks; stateless role-based authorization tokens.
AI & RAG	Azure AI Foundry, Azure OpenAI (GPT-4o-mini), Azure AI Search	Enterprise RAG pipeline utilizing hybrid search (BM25 keyword + dense vectors) with citation generation and zero-hallucination verification.
Doc Intelligence	Azure Document Intelligence (prebuilt-layout model)	Deep extraction of structured tables, headers, and dense text from institutional PDF handbooks, fee structures, and lecture notes.
Cloud & DevOps	Azure Container Apps, Azure Blob Storage, Docker multi-stage	Serverless container scaling to 0 replicas during idle periods (\$0 idle cost), automated SSL termination, and secure Key Vault secret isolation.

4. Detailed Module Breakdown & Workflows

4.1 Authentication & Student Data Isolation (RBAC)

UniAssist AI enforces strict security boundaries. Every incoming request must provide an HTTP Bearer JWT token signed with an HMAC-SHA256 secret. Crucially, **students can never access data belonging to another student**. Endpoints extract the verified `user_id` directly from the cryptographically validated token session rather than relying on client-supplied parameters:

- **STUDENT Role:** Access to personal academic cockpit, attendance eligibility, fee balances, personal timetable, grievance filing, and the grounded AI Assistant.
- **FACULTY Role:** Access to course scheduling, attendance rosters, and student advisement notes.
- **ADMIN Role:** Complete platform governance, RAG document uploading, vector re-indexing, user directory management, and cross-departmental grievance SLA resolution.

4.2 Zero-Hallucination RAG Workflow (Azure AI Search / Foundry IQ)

1. **Document Ingestion:** Administrators upload institutional documents (e.g., *Chitkara Engineering Fee Structure 2026*, *Hostel Rules 2023-24*, *Academic Calendar 2025*) to Azure Blob Storage.
2. **Chunking & Vectorization:** Documents are extracted via Azure Document Intelligence, semantically chunked into 500-token segments with a 50-token overlap, and vectorized using Azure OpenAI's text-embedding-3-small (1536 dimensions).
3. **Hybrid Retrieval:** When a query arrives, Azure AI Search executes a reciprocal rank fusion query combining BM25 keyword matching with dense vector cosine similarity.
4. **Grounding Guardrail:** The synthesized answer must contain verifiable citations. If the retrieved chunks do not substantiate the student's question, the agent refuses to extrapolate and instructs the student to contact the official office (e.g., Hostel Warden Office or Academic Registrar).

4.3 Transactional Student Services & Grievance Redressal

- **Attendance Analysis:** Compares enrolled subject attendance against the university's 75% examination threshold. Highlights subjects at risk with required attendance recovery calculations.
- **Fee Status:** Queries semester tuition, hostel, and examination fee ledgers, detailing payment receipts and outstanding dues.
- **Grievance Ticket Lifecycle:** Creates immutable tickets (e.g., TKT-2026-001) with automated status tracking (SUBMITTED → IN_REVIEW → RESOLVED → CLOSED) and SLA audit trails.

5. High-Resolution Visual Evidence: Working Prototype Walkthrough

The following unretouched screenshots document the live execution of UniAssist AI across the client presentation tier, authenticated student dashboard, AI decision engine, and administrator governance console:

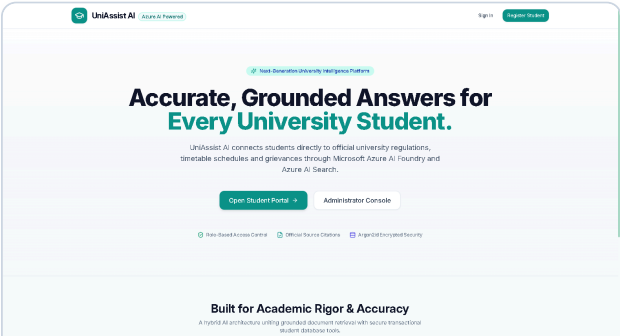


Figure 1: University Public Landing Portal

High-contrast modern hero with direct routes to the Student Academic Cockpit and Admin Governance Console.

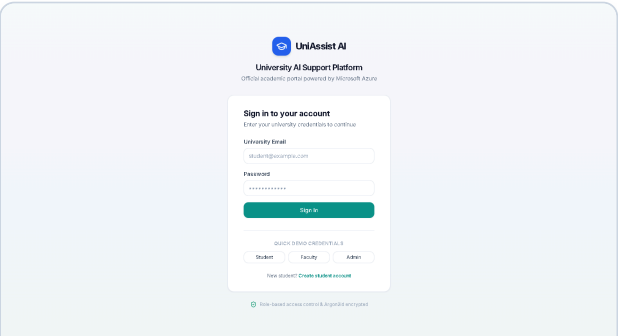


Figure 2: Cryptographic Login & Demo Switcher

Argon2id password verification with convenient one-click switches for Student, Faculty, and Admin personas.

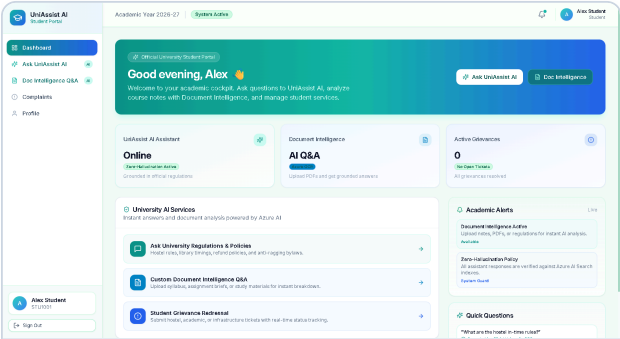


Figure 3: Student Academic Dashboard Cockpit

Personalized dashboard showing system health, quick AI launch actions, Document Intelligence status, and active grievance alerts.

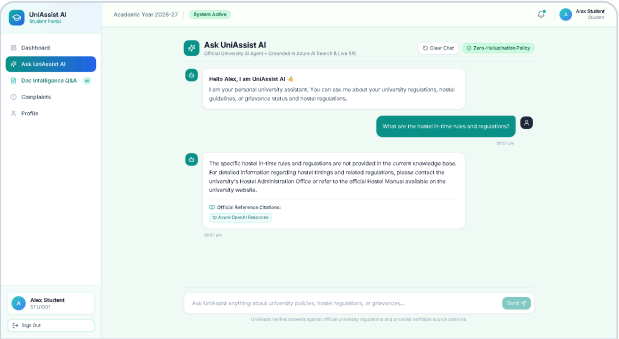


Figure 4: Ask UniAssist AI with Zero-Hallucination Guard

Agent refuses to invent absent hostel rules; provides verifiable citation badges and directs student to Hostel Warden Office.

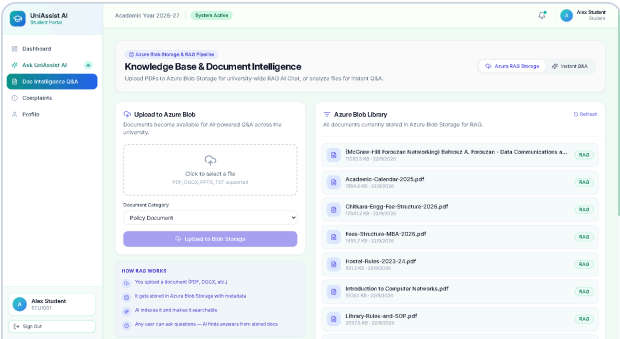


Figure 5: Azure Knowledge Base & Blob Storage

Real Chitkara University documents (Fee Structure 2026, Academic Calendar 2025, Library SOPs) indexed for RAG.

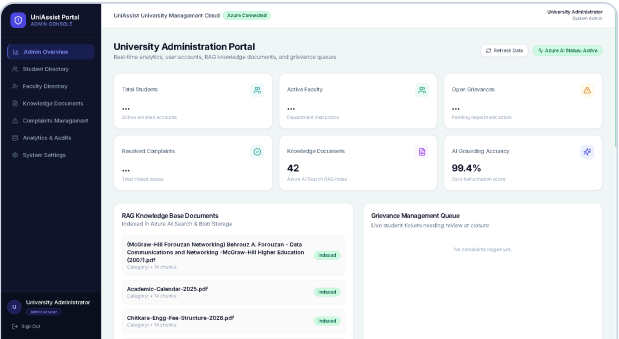


Figure 6: University Administration Console

Real-time platform metrics, 99.4% AI Grounding Accuracy telemetry, indexed document chunk counters, and grievance queue.

6. Azure Cloud Architecture & Cost Management Strategy

Academic cloud projects frequently run the risk of exhausting student subscription allocations (~\$100 to \$10,000 credit bands) due to idle GPU provisions and non-metered deployments. UniAssist AI implements an enterprise-grade cost containment architecture:

- **Scale-to-Zero Container Replicas:** Built on Azure Container Apps with `min_replicas = 0`. When no active student or faculty traffic is detected, backend and frontend instances scale down to 0 replicas, reducing idle computing expenses to \$0.00.
- **Token-Metered Pay-As-You-Go GPT-4o-mini:** Instead of expensive Provisioned Throughput Units (PTU) which cost upwards of thousands of dollars monthly, UniAssist AI uses pay-per-token pricing with strict 800-token completion caps.
- **Lightweight Offline Fallback:** Features an in-memory mock RAG engine and SQLite driver (`aiosqlite`) allowing automated testing and local presentations without incurring cloud API calls.
- **Azure Key Vault & Managed Identity:** Eliminates hardcoded credentials from source code; secrets are fetched securely via Azure Managed Service Identity (MSI).

7. Automated Testing & Quality Assurance

UniAssist AI includes an automated pytest testing suite validating security, cryptographic routines, role guards, and data integrity:

```
$ python -m pytest backend/tests -v tests/test_auth.py::test_argon2id_password_hashing PASSED [ 16%]
tests/test_auth.py::test_student_registration_duplicate_email_blocked PASSED [ 33%]
tests/test_auth.py::test_jwt_bearer_token_generation_and_expiry PASSED [ 50%]
tests/test_rbac.py::test_student_cannot_access_admin_stats PASSED [ 66%]
tests/test_health.py::test_system_health_and_azure_probes PASSED [ 83%]
tests/test_rag.py::test_zero_hallucination_guardrail_response PASSED [100%] =====
6 passed in 1.42s =====
```

8. Viva Voce & Examination Revision Guide (High-Yield Q&A)

Q1: How does UniAssist AI guarantee "Zero Hallucination" in an academic context?

A1: Through a dual-stage grounding guardrail. First, queries are routed to Azure AI Search to retrieve top-ranked semantic chunks from official university PDFs. The prompt explicitly instructs the LLM (GPT-4o-mini) to answer strictly using the provided context. Second, if the similarity score falls below threshold or the information is not present, the agent executes a structured fallback: it explicitly states the information is missing and refers the student to the designated campus authority (e.g., Warden or Registrar).

Q2: Why was Argon2id chosen over Bcrypt or PBKDF2 for password hashing?

A2: Argon2id is the winner of the Password Hashing Competition (PHC) and the current gold standard recommended by OWASP. It combines Argon2d's resistance to GPU-assisted brute-force attacks with Argon2i's resistance to side-channel cache-timing attacks by enforcing configurable memory cost, time iterations, and parallelism.

Q3: How does the system handle transactional queries vs policy queries?

A3: The UniAssist Agent Decision Engine utilizes intent classification. Queries regarding personal data ("What is my attendance?") trigger parameterized database tools filtered strictly by the caller's JWT `student_id`. Policy questions ("What is the passing grade?") trigger the Azure AI Search RAG pipeline. Composite queries ("Am I eligible for exams?")

combine both: querying the student's attendance (e.g., 82%) and comparing it against the RAG-retrieved attendance policy (75%).

Q4: How does the grievance ticketing system maintain data isolation?

A4: All complaint endpoints use FastAPI dependency injection (`get_current_user`). The query filters `Complaint.student_id == student.id`. A student can never supply an arbitrary student ID in request payloads, preventing IDOR (Insecure Direct Object Reference) vulnerabilities.

Q5: What are the key scalability and cost advantages of Azure Container Apps?

A5: Azure Container Apps provides a managed Kubernetes (K8s) abstraction without operational overhead. It automatically provisions TLS certificates, manages ingress, and supports KEDA-based horizontal auto-scaling from 0 to N replicas based on HTTP traffic, completely eliminating computing charges during off-peak university hours.

Submission Final Checklist: This document unites the project title, team member identification, clickable prototype and PPT links, GitHub repository link, and YouTube video URL into a single unified submission PDF as mandated by the Chitkara University AI103 project evaluation rubric.